

LECTURE #2

Econometrics I

SIMPLE REGRESSION MODEL

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In the previous lecture #1

- ▶ We introduced econometrics as a discipline and talked about causality and ceteris paribus concepts.
- ▶ We briefly reviewed basic statistical concepts: random variables, probability distributions, and their features.
- ▶ We refreshed selected distributions (normal, χ^2 , t , F).
- ▶ We recapitulated the intuition beyond LLN and CLT.
- ▶ We briefly summarized hypotheses testing.
- ▶ Readings for lecture #2:
 - ▶ Chapter 1: 1.3, Appendix C.1,
 - ▶ Chapter 2: 2.1, 2.2, **2.4 (mandatory for/after seminars)**.

Outline

Data structures

Simple regression model definition

Deriving the OLS

- Method of moments estimation

- Least squares estimation

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Data structures

There are, in general, four types of data analyzed in econometrics:

- ▶ cross-sectional data,
- ▶ time series data,
- ▶ independently pooled cross-sectional data,
- ▶ panel (longitudinal) data.

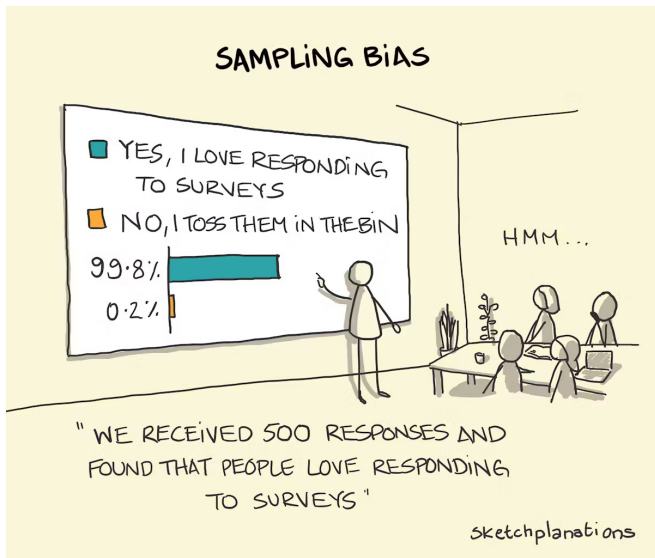
Cross-sectional data

- ▶ Sample of individuals, households, companies, countries, etc.
at a given time.
- ▶ Randomly sampled.
- ▶ Microeconomics, labor economics, public finance, health economics, industrial organization, etc.

Random sampling violated: Sampling biases

- ▶ **Selection bias:** occurs when the sample selection systematically excludes or underrepresents certain groups.
 - ▶ example: opinion poll about tuition payments among undergraduate students vs. all citizens.
- ▶ **Self-selection bias:** occurs when we examine data for a group of people who 'have chosen' to be in that group.
 - ▶ example: accident statistics of people who buy collision insurance ('convenience sample').
- ▶ **Survivor bias:** occurs when we choose a sample from a current population (survivors) in order to draw inferences about a past population.
 - ▶ example: S&P500 companies, medical records of older people.
- ▶ **Nonresponse bias:** the systematic refusal of some groups to participate in an experiment or to respond to a poll.
 - ▶ example: expats.

Random sampling violated: What bias is this?



Source: sketchplanations.com [2025-01]

Time series data

- ▶ Observations on a variable or a set of variables **over time**.
- ▶ Most severe issue: hardly ever independent in time.
- ▶ Mainly macroeconomics, finance, and connected disciplines.
- ▶ Data frequency!

Independently pooled cross-sectional data

- ▶ Somewhere between cross-sectional and time series data but not perfectly structured.
- ▶ Data drawn from **different independent samples** collected in **different points in time**.
- ▶ E.g., observations from four different years with different randomly sampled cross-sectional groups for each year.
- ▶ Usually used to increase the number of observations or to analyze the effects of a specific policy in time.

Panel data

- ▶ Also called longitudinal data.
- ▶ **Time series** for **each cross-sectional unit** in the dataset.
- ▶ Cross-sectional units thus **remain the same** over time!
- ▶ Might be very difficult to collect.

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Simple regression model

- ▶ For variables x and y , the simple linear regression model is defined as

$$y = \beta_0 + \beta_1 x + u.$$

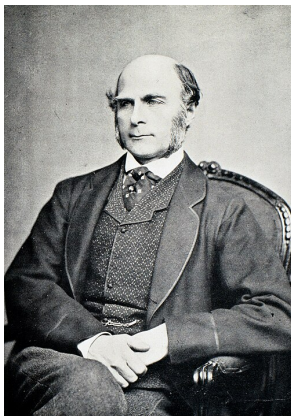
- ▶ Variables x and y are randomly sampled and stored in a dataset.
- ▶ **Parameters**/coefficients β_0 and β_1 are of interest as they represent and describe the relationship between x and y : 'how y varies with changes in x ' $\Rightarrow$ they need to be estimated.
- ▶ β_0 is called the **intercept parameter** or the constant term.
- ▶ β_1 is called the **slope parameter**.
- ▶ **Error term** or disturbance u represents unobserved effects that are not included in the regression equation.

Francis Galton

From Wikipedia, the free encyclopedia

"Galton" redirects here. For other uses, see [Galton \(disambiguation\)](#).

Sir Francis Galton, FRS (/ˈfrɑːnsɪs ˈɡɔːltən/; 16 February 1822 – 17 January 1911) was an English Victorian statistician, progressive, polymath, sociologist, psychologist,^{[1][2]} anthropologist, eugenicist, tropical explorer, geographer, inventor, meteorologist, proto-geneticist, and psychometrician. He was knighted in 1909.



Galton, Francis (1885): Opening address, Section H, Anthropology, Nature, Sept. 24, pp. 507

just.

The subject of my remarks will be "Types and their Inheritance." I shall discuss the conditions of the stability and instability of types, and hope in doing so to place beyond doubt the existence of a simple and far-reaching law that governs hereditary transmission, and to which I once before ventured to draw attention, on far more slender evidence than I now possess.

It is some years since I made an extensive series of experiments on the produce of seeds of different size but of the same species. They yielded results that seemed very noteworthy, and I used them as the basis of a lecture before the Royal Institution on February 9, 1877. It appeared from these experiments that the offspring did *not* tend to resemble their parent seeds in size, but to be always more mediocre than they—to be smaller than the parents, if the parents were large; to be larger than the parents, if the parents were very small. The point of convergence was considerably below the average size of the seeds contained in the large bagful I bought at a nursery-garden, out of which I selected those that were sown.

The experiments showed further that the mean filial regression towards mediocrity was directly proportional to the parental deviation from it. This curious result was based on so many plantings, conducted for me by friends living in various parts of the country, from Nairn in the north to Cornwall in the south, during one, two, or even three generations of the plants, that I could entertain no doubt of the truth of my conclusions. The

Simple regression model

- ▶ For variables x and y , the simple linear regression model is defined as

$$y = \beta_0 + \beta_1 x + u.$$

- ▶ Population vs. sample.
- ▶ Variables x and y are usually referred to as

y	x
Dependent variable Explained variable Response variable Predicted variable Regressand	Independent variable Explanatory variable Control variable Predictor variable Regressor

- ▶ Linear functional form implies a linear ceteris paribus effect between x and y

$$\Delta y = \beta_1 \Delta x \quad \text{if} \quad \Delta u = 0.$$

Linearity of the model

- ▶ Regression model from the previous slide is also referred to as the **linear regression**.
- ▶ Linearity in parameters β_k , $k = \{0, 1\}$.
- ▶ This does not limit the construction of variables.
- ▶ However, the interpretation of parameters changes.

Logarithmic functional forms

- ▶ Some relationships are not linear by their nature.
- ▶ For interpretational purposes, it is sometimes more appropriate or desirable to comment on the percentage effects.
- ▶ Logarithmic transformations are popular from the perspective of economics as they allow for elasticity and semi-elasticity interpretations:

Model	Dependent v.	Independent v.	Interpretation of β_1
Level-level	y	x	$\Delta y = \beta_1 \Delta x$
Level-log	y	$\log(x)$	$\Delta y = (\beta_1/100)\% \Delta x$
Log-level	$\log(y)$	x	$\% \Delta y = (100\beta_1) \Delta x$
Log-log	$\log(y)$	$\log(x)$	$\% \Delta y = \beta_1 \% \Delta x$

Our objective

- ▶ Estimate the 'betas'!

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

- ▶ Do so with statistical precision, i.e., to know the estimator's bias, variance, or consistency characteristics and related asymptotic properties.
- ▶ Deal with possible violations of important assumptions.
- ▶ Deal with different types of data and possible subpopulations.

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Expected value of the error term

- ▶ As long as there is the intercept β_0 in the model, we can assume that

$$\mathbb{E}(u) = 0. \quad (1)$$

- ▶ This is rather a statement than an assumption, as we can always make it hold.

Relationship between x and u

- ▶ Even though correlation might be appealing, it does not control for large enough set of possible relationships: e.g., x vs. x^2 vs. $|x|$.
- ▶ It is much more appropriate to use conditional expectations.
- ▶ Specifically, we assume that

$$\mathbb{E}(u|x) = \mathbb{E}(u), \quad (2)$$

i.e., u is **mean independent** of x .

- ▶ Combining (1) and (2), we obtain the **zero conditional mean assumption**

$$\mathbb{E}(u|x) = 0.$$

- ▶ This also gives us the **population regression function (PRF)**

$$\mathbb{E}(y|x) = \beta_0 + \beta_1 x.$$

- ▶ Mind the difference between $\mathbb{E}(y|x)$ and y here.

Two important (implied) equations

This leads to two practically important (implied) equations and motivations that we can use for the estimation of parameters β_0 and β_1 :

$$\mathbb{E}(u) = 0,$$

$$\text{Cov}(x, u) = \mathbb{E}(xu) - \mathbb{E}(x)\mathbb{E}(u) = \mathbb{E}(xu) = 0.$$

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Method of moments estimation

- ▶ Method of moments (MM) estimation stems in specifying moments' restrictions and substituting them with their sample counterparts based on n observations in the dataset.
- ▶ In our case, we take the assumptions about the unconditional expected value of u and covariance between x and u and rewrite them as

$$\mathbb{E}(u) = \mathbb{E}(y - \beta_0 - \beta_1 x) = 0,$$

$$\text{Cov}(x, u) = \mathbb{E}(x(y - \beta_0 - \beta_1 x)) = 0.$$

- ▶ Rewriting the expectations as the sample moments, we get the estimators by solving

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0,$$

$$\frac{1}{n} \sum_{i=1}^n (x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)) = 0.$$

Method of moments estimation

- ▶ For clarity, unless stated otherwise, $\sum := \sum_{i=1}^n$.
- ▶ From the first one, we get

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 &\iff \frac{1}{n} \sum y_i - \frac{1}{n} \hat{\beta}_0 \sum 1 - \frac{1}{n} \hat{\beta}_1 \sum x_i = 0 \iff \\ \bar{y} - \hat{\beta}_0 - \frac{1}{n} \hat{\beta}_1 n \bar{x} = 0 &\iff \boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}.\end{aligned}$$

- ▶ From the second one, we get (substitute for $\hat{\beta}_0$ and solve for $\hat{\beta}_1$)

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n (x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)) = 0 &\iff \sum x_i (y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i) = 0 \iff \\ \sum x_i (y_i - \bar{y}) - \hat{\beta}_1 \sum x_i (x_i - \bar{x}) = 0 &\iff \boxed{\hat{\beta}_1 = \frac{\sum x_i (y_i - \bar{y})}{\sum x_i (x_i - \bar{x})}}.\end{aligned}$$

Method of moments estimation

- ▶ We thus get to

$$\hat{\beta}_1^{OLS} = \frac{\sum_{i=1}^n x_i(y_i - \bar{y})}{\sum_{i=1}^n x_i(x_i - \bar{x})}.$$

- ▶ Here, we assume that the denominator is nonzero (even, more specifically, positive).
- ▶ As a byproduct, we also obtained

$$\hat{\beta}_0^{OLS} = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- ▶ Estimator vs. estimate.
- ▶ Finally, for a given sample of data, we can express the estimated model numerically

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

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- ▶ Let us first define a **fitted value** for y when $x = x_i$ as

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

- ▶ **Residual** is defined as

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

- ▶ Specifying the **sum of squared residuals**

$$SSR \equiv \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

as a penalty function of interest, we get the same estimators $\hat{\beta}_0^{OLS}$ and $\hat{\beta}_1^{OLS}$ as for MM.

- ▶ Hence the name OLS: 'Ordinary Least Squares' estimator.

Least squares estimation

- ▶ We have: $SSR \equiv \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$.
- ▶ Looking for the minimum of the penalty function, we take partial derivatives with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$, i.e., FOCs

$$\frac{\partial SSR}{\partial \hat{\beta}_0} = 0 \quad \Longleftrightarrow \quad -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0,$$

$$\frac{\partial SSR}{\partial \hat{\beta}_1} = 0 \quad \Longleftrightarrow \quad -2 \sum_{i=1}^n (x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)) = 0.$$

- ▶ These are exactly comparable to the MM derivation, i.e., they lead to the same solution.
- ▶ Alternative notations of $\hat{\beta}_1^{OLS}$ are given as

$$\hat{\beta}_1^{OLS} = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})} = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \boxed{\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$

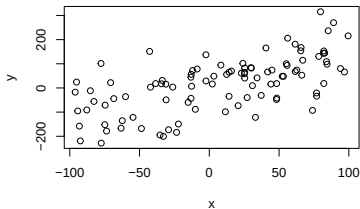
OLS regression line and additional notes

- ▶ Parallel to the population regression function, we also get the **sample regression function (SRF)** or the **OLS regression line**:

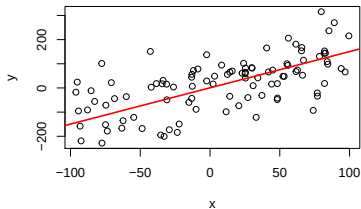
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

- ▶ Note the essential differences:
 - ▶ parameter vs. estimator vs. estimate,
 - ▶ observations vs. expected values vs. fitted values,
 - ▶ population regression function (PRF) is fixed for the population but unknown,
 - ▶ in general, the PRF and SRF differ,
 - ▶ and for each sample of data, the SRF (OLS regression line) differs as well.
- ▶ We say that ‘we run the regression of y on x ,’ or simply ‘we regress y on x ’ or even more simply ‘ $y \rightarrow x$.’

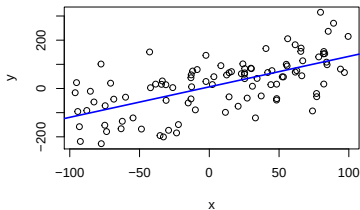
(a) $y = 0.5 + 1.5x + u$, $n = 100$



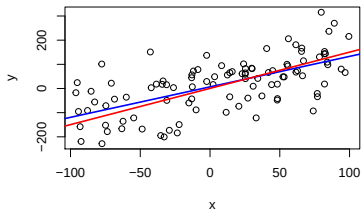
(b) PRF: $\mathbb{E}(y|x) = 0.5 + 1.5x$



(c) SRF (OLS RL): $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$



(d) SRF $\neq$ PRF



Seminars and the next lecture

- ▶ Seminars:
 - ▶ interpretation of estimated simple regression models,
 - ▶ discussion on ceteris paribus and causal effects,
 - ▶ predictions,
 - ▶ computer exercise with empirical data (BYOD?),
 - ▶ changing units of measurement and logs (TAKE HOME, **2.4**).
- ▶ Next lecture #3:
 - ▶ basic OLS properties (algebraic, R^2),
 - ▶ expected values and variances of the OLS estimators
 - ▶ unbiasedness,
 - ▶ variance,
 - ▶ regression through the origin.
- ▶ Readings for lecture #3:
 - ▶ Chapter 2: 2.3. 2.5, **2.6 (mandatory for/after seminars)**.

Appendix: Various forms of parentheses in the sum

$$\hat{\beta}_1^{OLS} = \frac{\sum_{i=1}^n x_i(y_i - \bar{y})}{\sum_{i=1}^n x_i(x_i - \bar{x})} = \frac{\sum_{i=1}^n y_i(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \boxed{\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

- Using $\sum x_i = n\bar{x}$ and $\sum y_i = n\bar{y}$, you can expand the numerator

$$\boxed{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})} = \sum (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) = \\ = \sum x_i y_i - \bar{y} \sum x_i - \bar{x} \sum y_i + n\bar{x}\bar{y} = \sum x_i y_i - n\bar{x}\bar{y}.$$

- Finally, $n\bar{x}\bar{y}$ can be written both as $\bar{y} \sum x_i$ or $\bar{x} \sum y_i$, which gives us two options how to write the original sum
 - $\sum x_i y_i - \bar{y} \sum x_i = \sum (x_i y_i - \bar{y} x_i) = \underline{\underline{\sum x_i (y_i - \bar{y})}}$
 - $\sum x_i y_i - \bar{x} \sum y_i = \sum (x_i y_i - \bar{x} y_i) = \underline{\underline{\sum y_i (x_i - \bar{x})}}$: L #3, 4, 5.

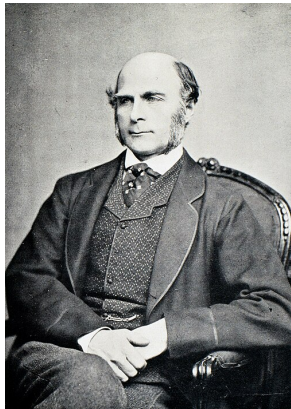
- Derivation for the denominator $\boxed{\sum (x_i - \bar{x})^2} = \sum x_i (x_i - \bar{x})$ is, actually, the same.

Historical spotlight: Francis Galton (1822–1911)

Francis Galton gave econometrics two day-one tools: correlation and the regression line. He first saw what he called 'regression toward mediocrity' in sweet-pea seeds, comparing parent seeds' size with their offspring. He presented the idea in 1877, and his *Typical laws of heredity* was published in *Nature*.

He then extended the analysis to human body height. Using thousands of measurements from his Anthropometric Laboratory, he wrote *Regression towards mediocrity in hereditary stature* (1886): tall parents have tall children on average, but closer to the mean; short parents have short children, but also nearer the mean. Soon after came *Co-relations and their measurement* (1888), which showed how to quantify relationship between characteristics. Galton thus coined the term 'co-relation,' used scatter plots and standard deviations to compare variables, and linked line slope to relationship strength, laying the groundwork for Pearson's correlation. That is why we speak of regression lines and correlation today.

Why it matters: without regression toward the mean, repeated selection would split humanity into subpopulations of 'giants and dwarfs,' which we do not observe. Implications for econometric is clear: linear regression characterizes the average tendency, not a deterministic rule for individuals, so predictions for extreme cases require caution.



Source: [Wikipedia](#)